DIPLOMA ELECTRONICS / ECE / BIOMEDICAL - SEMESTER 3 LAB

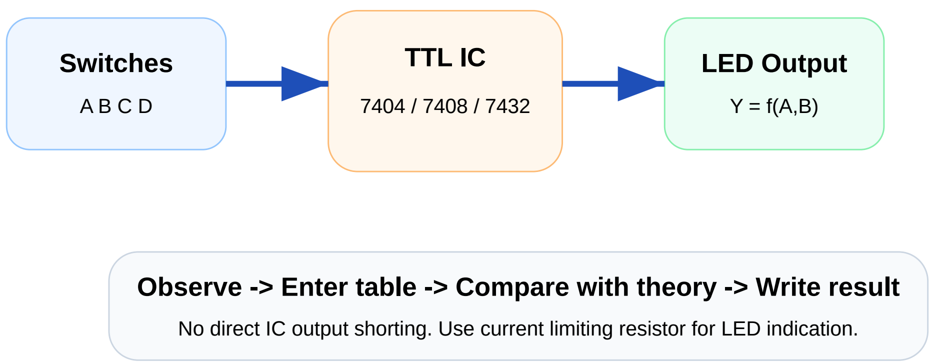
Digital Electronics Lab

A practical handbook for Course Code 3048. This is designed like a complete lab manual: aim, apparatus, IC identification, theory, wiring method, truth tables, observation templates, result writing, common faults, viva questions and model lab examination.

COURSE CODE	**3048**
CREDITS	**1.5**
PERIODS	**45 practical hours**
CATEGORY	**Program Core**

Lab bench view

Input switches -> TTL gates -> output LEDs -> truth table -> result. Every experiment must prove the circuit, not only connect wires.



General workflow used for all Digital Electronics Lab experiments.

OFFICIAL STRUCTURE

Syllabus Map and Learning Outcomes

The syllabus is practical-only: 3 periods per week, 45 periods per semester, with 6 hours for basic TTL gates, 12 hours for combinational circuits, 9 hours for flip-flops/shift registers, 15 hours for counters and 3 hours for lab examination.

Objectives

- ✓ Reinforce theoretical Digital Electronics through hardware practice.
- ✓ Implement simple digital circuit based applications.
- ✓ Develop correct wiring, observation and troubleshooting habits.

Prerequisites

Electric circuits and components from Fundamentals of Electrical and Electronics Engineering Lab; combinational and sequential circuits from Digital Electronics.

Professional use

Digital logic is the base of elevator/escalator controller boards, display boards, sensor logic, encoder inputs and safety interlock checking.

CO	Outcome	Hours	Core skill
CO 1	Develop basic logic functions using TTL logic gates.	6	Truth-table verification and universal gate use
CO 2	Construct combinational logic circuits using logic gates.	12	Adders, code converters, MUX and DEMUX
CO 3	Construct flip-flops and shift registers using gates and ICs.	9	Sequential memory and controlled shifting
CO 4	Construct synchronous and asynchronous counters using flip-flop ICs.	15	Counter sequence, reset logic and 7-segment display

Do not treat this as a wiring-only lab

If the output LED glows, that alone is not a result. A valid lab result needs: correct IC selection, power pin check, input state table, measured/observed output, comparison with expected output, reason for mismatch and final result statement.

□□□□□ wire connect □□□□□□□□ □□□□□□□ □□□□. Truth table prove □□□□□□□□. Output expected value-□□□□ match □□□□□□□□□□□□ □□□ □□□□□□□ result.

LAB DISCIPLINE

Standard Lab Record Format

Use this same structure for every experiment. This avoids incomplete records and helps in internal evaluation.

Record page order

1. Experiment number, date, title and CO mapping.
2. Aim written in one clear sentence.
3. Apparatus/components: trainer kit, IC numbers, resistor values, LEDs, jumpers, power supply.
4. Theory: short logic explanation and equation.
5. IC pin diagram or important pins.
6. Circuit diagram / connection table.
7. Truth table or observation table.
8. Procedure, precautions, result and viva answers.

Before switching ON

- ✓ Confirm +5 V and GND rails; TTL ICs are not operated directly from 9 V or 12 V.
- ✓ Check IC notch direction. Wrong IC orientation can damage ICs.
- ✓ Never leave TTL inputs floating. Tie unused inputs to valid logic 0 or 1 as required.
- ✓ Use current-limiting resistors with LEDs unless the trainer kit already includes them.
- ✓ Switch off supply before changing major wiring.

Universal observation table template

Sl.No	Input A	Input B	Input C / Mode	Expected Output	Observed Output	Remarks
1	0	0	-	-	-	Match / mismatch
2	0	1	-	-	-	Match / mismatch
3	1	0	-	-	-	Match / mismatch
4	1	1	-	-	-	Match / mismatch

CO1 - 6 HOURS

Module 1: Basic Logic Functions using TTL Gates

Experiments: verify NOT, OR, AND, NAND, NOR, XOR and XNOR truth tables; implement any logic function using NAND and NOR gates only.

Experiment 1A - Verify TTL logic gates

Aim: Verify the truth tables of NOT, OR, AND, NAND, NOR, XOR and XNOR gates using TTL ICs.

Typical ICs: 7404 NOT 7432 OR 7408 AND 7400 NAND 7402 NOR 7486 XOR 74266 XNOR

Basic rule: Pin 14 = +5 V and Pin 7 = GND for most 14-pin TTL ICs used here.

Truth table summary

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

How to write result

The truth table of the given TTL logic gate was verified. Observed outputs matched expected outputs for all possible input combinations.

> Output LED ON = logic 1, OFF = logic 0 □□□□□□ clear □□□ record □□□□□□□□. Trainer kit active-low LED □□□□□□□□□ note □□□□□□□.

Procedure

1. Place the required IC in the trainer kit or breadboard with notch direction noted.
2. Connect +5 V to VCC and GND to ground pin.
3. Connect input switches to input pins through trainer connections.
4. Connect output pin to LED indication point.
5. Apply all input combinations and enter observed output.
6. Compare with theoretical truth table and conclude.

Common mistakes

- Forgetting VCC/GND, then assuming IC is faulty.
- Connecting output of one gate directly to another output. Outputs must not be shorted together.
- Using NOR IC pinout like NAND IC pinout. Different TTL ICs can have different internal gate pin positions.
- Leaving unused input floating and getting unstable output.

Experiment 1B - Universal gates: NAND-only and NOR-only realization

Aim: Develop any logic function using NAND gates only and NOR gates only. NAND and NOR are called universal gates because NOT, AND and OR can be built from them.

NAND equivalents

NOT A = A NAND A

A AND B = (A NAND B) NAND (A NAND B)

A OR B = (A NAND A) NAND (B NAND B)

NOR equivalents

NOT A = A NOR A

A OR B = (A NOR B) NOR (A NOR B)

A AND B = (A NOR A) NOR (B NOR B)

Example function: F = A + B.C. First build B.C, then OR with A. In NAND-only form use De Morgan: F = ((A)' . (B.C)')' and implement using NAND stages.

CO2 - 12 HOURS

Module 2: Combinational Logic Circuits

Experiments: half adder, full adder, binary-gray conversion, 4:1 MUX, 1:4 DEMUX, MUX IC 74151 and Decoder/Demux IC 74155.

Experiment 2A - Half Adder and Full Adder

Half Adder: Adds two one-bit inputs A and B.

Sum = A XOR B, Carry = A.B

Full Adder: Adds A, B and carry input Cin.

Sum = A XOR B XOR Cin

Carry = A.B + B.Cin + A.Cin

A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Experiment 2B - Binary to Gray and Gray to Binary

Binary to Gray: MSB remains same. Each next Gray bit is XOR of adjacent binary bits.

G3=B3, G2=B3 XOR B2, G1=B2 XOR B1, G0=B1 XOR B0

Gray to Binary: MSB remains same. Each next binary bit is XOR of previous binary bit and current gray bit.

B3=G3, B2=B3 XOR G2, B1=B2 XOR G1, B0=B1 XOR G0

Gray code-□ adjacent values □□□□□□□ □□□ bit □□□□□□□ change □□□□□□□□. Sensor/encoder systems-□ false transition □□□□□□□□□□□ □□□ □□□□□□□□□□□□□□.

Experiment 2C - 4:1 Multiplexer using gates

A multiplexer selects one of many inputs and sends it to one output. For 4:1 MUX, two select lines S1 and S0 select D0, D1, D2 or D3.

Y = D0.S1'.S0' + D1.S1'.S0 + D2.S1.S0' + D3.S1.S0

S1	S0	Output Y
0	0	D0
0	1	D1
1	0	D2
1	1	D3

Experiment 2D - 1:4 De-multiplexer

A de-multiplexer sends one input data line to one of many output lines using select inputs.

Y0=D.S1'.S0', Y1=D.S1'.S0, Y2=D.S1.S0', Y3=D.S1.S0

For testing, keep D=1 and vary select lines. Only one output should become active for each select state.

Experiment 2E - 74151 MUX and 74155 Decoder / Demux

IC 74151 is an 8:1 multiplexer. IC 74155 is a dual 2-line to 4-line decoder/demultiplexer. In lab, learn the enable pins carefully because many TTL decoder outputs are active-low. A wrong enable state makes every output look wrong even when input wiring is correct.

- ✓ Read the IC notch and pin numbering before wiring.
- ✓ Connect enable/strobe pins to the required logic level.
- ✓ Use select lines systematically and write outputs in the observation table.
- ✓ For active-low output ICs, mention that LED ON/OFF relation may be inverted depending on trainer kit.

CO3 - 9 HOURS

Module 3: Flip-Flops and Shift Registers

Experiments: SR flip-flop using NAND gates, conversion to D flip-flop, demonstration of IC 7474 and IC 7476, controlled shift register with left/right mode control.

Experiment 3A - SR Flip-Flop using NAND Gates

An SR latch stores one bit. In a NAND latch the inputs are usually active-low S' and R'.

S'	R'	Q(next)	Meaning
0	1	1	Set
1	0	0	Reset
1	1	Previous Q	Hold
0	0	Invalid	Forbidden

D conversion: Feed S = D and R = D' through suitable gating so that forbidden input is avoided.

Experiment 3B - D Flip-Flop IC 7474

7474 is a dual positive-edge triggered D flip-flop with preset and clear. Preset and clear are asynchronous and commonly active-low.

- ✓ Keep PRE and CLR inactive during normal clocked operation.
- ✓ Apply D input, then give clock edge and observe Q.
- ✓ Q follows D only on the triggering clock edge, not continuously.

> At active clock edge: Q(next) = D

Experiment 3C - JK Flip-Flop IC 7476

JK flip-flop removes the invalid condition of SR flip-flop. When J=K=1, output toggles on the clock event.

J	K	Q(next)	Operation
0	0	Q	Hold
0	1	0	Reset
1	0	1	Set
1	1	Q'	Toggle

Experiment 3D - Controlled Shift Register

A shift register moves stored data left or right with clock pulses. Mode control selects direction. This experiment proves sequential data movement, not just final LED pattern.

1. Clear all flip-flops.
2. Load starting data or serial input.
3. Select left/right mode.
4. Apply one clock at a time and record Q outputs after every clock.
5. Reverse mode and repeat observation.

> Clock pulse □□□ □□□ □□□□□□□□□□□□□□□ output pattern □□□□□□□. □□□□□□□ □□□□□□ □□□□□□□□□ shift direction prove □□□□□□□ □□□□□□□□.

CO4 - 15 HOURS

Module 4: Synchronous and Asynchronous Counters

Experiments: asynchronous decade counter 0-9 with 7-segment display, IC 7490 and IC 7493, synchronous up-down counter, Johnson counter and ring counter.

Experiment 4A - Asynchronous Decade Counter with 7-Segment Display

An asynchronous counter is also called ripple counter because clock reaches the next flip-flop from the previous flip-flop output. To make a decade counter, reset logic forces the count back to 0000 after 1001.

Decade count sequence: 0000, 0001, 0010, ... , 1001, reset to 0000

- ✓ Use clean clock pulses; switch bounce causes skipped counts.
- ✓ Check reset gate connections carefully.
- ✓ Use BCD to 7-segment driver if trainer kit does not contain internal decoder.

Experiment 4B - IC 7490 and IC 7493

7490: Decade/BCD counter commonly used for divide-by-10 operation.

7493: 4-bit binary ripple counter commonly used for divide-by-16 operation.

Important practical point: Reset pins decide the actual count mode. If reset pins are left wrongly connected, the IC will not count in the expected sequence.

Experiment 4C - Synchronous Up-Down Counter

In a synchronous counter, all flip-flops receive the same clock. Logic gates decide whether each flip-flop toggles. This gives better speed than ripple counters.

- ✓ Mode UP: sequence increases with every clock.
- ✓ Mode DOWN: sequence decreases with every clock.
- ✓ Record at least one full cycle in both directions.
- ✓ Use 7-segment display to verify decimal count along with binary LEDs.

Experiment 4D - Ring and Johnson Counters

Ring counter: A single 1 circulates through the flip-flop chain. For 4 flip-flops: 1000 -> 0100 -> 0010 -> 0001 -> 1000.

Johnson counter: Inverted output of the last flip-flop is fed back to first input. A 4-bit Johnson counter gives 8 states.

Clock	4-bit Ring	4-bit Johnson
0	1000	0000
1	0100	1000
2	0010	1100
3	0001	1110
4	1000	1111

Counter result writing

The counter was constructed and tested. The observed output sequence followed the expected binary/BCD/ring/Johnson sequence for each clock pulse. Reset and mode control operation were verified.

BENCH REFERENCE

TTL IC Quick Reference and Pin Discipline

This section is for quick checking during lab. Always verify with actual IC datasheet/trainer manual before final wiring.

IC	Function	Typical use in this lab	Special caution
7400	Quad 2-input NAND	Universal gate, SR latch	Do not confuse with 7402 NOR pinout
7402	Quad 2-input NOR	NOR-only realization	Output pins differ from NAND IC
7404	Hex inverter	NOT and complement generation	Unused inverter inputs must not float
7408	Quad 2-input AND	Adders and decoder logic	Use with XOR for adder circuits
7432	Quad 2-input OR	Carry logic, MUX/DEMUX sum terms	Check output loading
7486	Quad XOR	Sum, Gray code conversion	XOR output is 1 only for unequal inputs
74151	8:1 MUX	Data selection demonstration	Enable/strobe pin is critical
74155	Dual decoder/demux	Demux / decoding	Many outputs are active-low
7474	Dual D flip-flop	Clocked storage, shift register	Preset/clear are asynchronous
7476	Dual JK flip-flop	Counters and toggling	Check clock edge and clear/preset
7490	Decade counter	0-9 BCD counter	Reset pins decide sequence
7493	Binary counter	Binary ripple counter	Reset to known state before test

Golden rules

1. Identify notch. 2. Count pins anticlockwise from notch. 3. Connect VCC/GND first. 4. Keep inputs valid. 5. Test one block at a time. 6. Record every state.

```
   notch
  ________
1|        |14 VCC
2|  TTL   |13
```

```
3|  IC    |12
4|        |11
5|        |10
6|________|9
7 GND      8
```

TROUBLESHOOTING

Fault Finding Guide

Most digital lab faults are wiring faults, not damaged ICs. Check logically before replacing components.

Output always 0

- VCC missing or GND missing.
- Output LED connected in reverse or active-low indication misunderstood.
- Gate enable input disabled.
- Input stuck at logic 0 due to switch or jumper fault.

Output always 1

- Input floating in TTL circuit.
- Pull-up in trainer board making input high.
- NAND/NOR active output logic misunderstood.
- Counter reset not released.

IC heating

- Supply voltage wrong or IC inserted reverse.
- Output shorted to output or directly to supply.
- Wrong pin connection to +5 V or ground.
- Remove supply immediately and inspect.

Counter skips counts

- Clock switch bounce.
- Reset gate glitch.
- Loose clock wire.
- Using asynchronous counter where propagation delay causes temporary states.

CONTINUOUS INTERNAL EVALUATION

Open-Ended Project Ideas

The syllabus suggests open-ended work as group activity. These are not for end semester examination, but must be used for continuous internal evaluation.

Project 1 - K-map simplification and gate implementation

Take a 3-variable or 4-variable Boolean function. Simplify it using K-map. Implement the simplified expression with logic gates and compare with the original truth table.

Deliverables

- ✓ Problem statement
- ✓ Truth table
- ✓ K-map grouping
- ✓ Simplified expression
- ✓ Circuit diagram
- ✓ Observation table and result

Project 2 - Counter with given specification

Design a counter such as MOD-6, MOD-10, odd-number counter, even-number counter, traffic-light sequence counter or machine-state sequence counter.

Deliverables

- ✓ Required sequence
- ✓ Flip-flop selection
- ✓ State table
- ✓ Reset/feedback logic
- ✓ Hardware verification
- ✓ Fault analysis

Industry-style project example

Escalator inspection counter demo: Create a counter that increments on each simulated sensor pulse and resets at a defined count. Display the count on 7-segment display. This connects Digital Electronics Lab to real control panel logic, display boards and sensor pulse counting.

QUESTION BANK

Viva and Oral Exam Questions

Basic gates

Q1 Why are NAND and NOR called universal gates?

Q2 What is the difference between XOR and OR?

Q3 Why should TTL inputs not be left floating?

Q4 What are VCC and GND pins in common 14-pin TTL ICs?

Q5 How do you identify pin number 1 of an IC?

Combinational circuits

Q6 Write sum and carry equations of half adder.

Q7 Why is full adder required in multi-bit addition?

Q8 What is the advantage of Gray code?

Q9 What is the function of select lines in MUX?

Q10 What is active-low output?

Sequential circuits

Q11 What is the difference between latch and flip-flop?

Q12 What is toggle condition in JK flip-flop?

Q13 Why are preset and clear called asynchronous inputs?

Q14 What is a shift register?

Q15 How is data shifted left or right?

Counters

Q16 What is ripple counter?

Q17 Difference between synchronous and asynchronous counter?

Q18 What is MOD number of a counter?

Q19 How does a decade counter reset after 9?

Q20 Difference between ring and Johnson counter?

Short answers

Show key answers

NAND/NOR are universal because any Boolean function can be realized using only NAND or only NOR. XOR output is high for unequal inputs. Gray code changes only one bit between adjacent values. A flip-flop is clock-controlled memory. Ripple counter has clock propagation through stages; synchronous counter clocks all stages together. MOD number is number of unique states in one counting cycle.

PRACTICE EXAM

Model Practical Lab Examination

This is a practice format for preparation. Follow your institution's actual lab exam instructions when given.

Model Task A - Logic Gate / Combinational

Task: Construct a full adder using logic gates and verify the truth table.

Expected record: Aim, IC list, equations, circuit connections, truth table, observed output and result.

Marks focus: Correct circuit, systematic wiring, all eight input combinations, proper result.

Model Task B - Sequential / Counter

Task: Construct a decade counter using suitable IC and display the count on 7-segment display.

Expected record: IC details, reset logic, count sequence table, display observation and result.

Marks focus: Reset operation, clock control, correct 0-9 sequence, safe wiring.

Suggested marking checklist

Area	Marks weight	What examiner checks
Circuit preparation	High	IC selection, pin identification, supply check, neat wiring
Execution	High	Correct input application and output observation
Record	Medium	Aim, theory, equation, table and result
Viva	Medium	Concept clarity and troubleshooting logic
Safety	Mandatory	No IC damage, no direct short, proper supply handling

Final revision checklist

- ✓ Memorize truth tables of all basic gates.
- ✓ Know NAND-only and NOR-only conversions.
- ✓ Practice full adder equations and Gray code conversion.
- ✓ Understand MUX/DEMUX select line operation.
- ✓ Know D and JK flip-flop operation tables.
- ✓ Know ring, Johnson, decade and binary counter sequences.
- ✓ Know IC supply pin convention and active-low reset/enable concept.

REFERENCES

Text, Reference and Online Resources

Books

- Digital Principles and Applications - Albert Paul Malvino and Donald P. Leach.
- Digital Electronics - Roger L. Tokheim.
- Digital Electronics: An Introduction to Theory and Practice - William H. Gothmann.
- Fundamentals of Logic Design - Charles H. Roth Jr.
- Digital Electronics - R. Anand.

Online learning

- Wikibooks Digital Electronics.
- GeeksforGeeks Digital Electronics and Logic Design tutorials.
- All About Circuits digital textbook.
- ResearchGate digital electronics material listed in syllabus.